Dharmenthiran T

Software Developer

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Summary

Software Developer with nearly 3 years of experience building cross-platform applications and backend services using **Flutter, Dart, Python**, and **FastAPI**. Experienced in developing REST APIs, industrial IoT applications, real-time communication systems using **MQTT and WebSockets**, and database integrations. Hands-on experience with **Docker, Nginx, CI/CD pipelines**, and application deployment. Strong understanding of software architecture, API security, and database management.

Education

Govt Arts And Science College (Autonomous) — Coimbatore, TN
Bachelor of Science in Information Technology • 72% — *2020 – 2023*

Chirumalar Higher Secondary School — Trichy, TN
Higher Secondary (HSC) • 74% — *2020*

Experience

Software Development Engineer I — Jan 2024 – Present
SPI Equipments India Pvt. Ltd. — *Coimbatore, TN*

- Developed and maintained cross-platform industrial applications using **Flutter** and **Dart** for Windows, Android, and macOS.
- Designed and implemented REST APIs and backend services using **Python** and **FastAPI**.
- Integrated industrial hardware using **TCP/IP**, **RS-232/RS-485**, **MQTT**, and **WebSocket** communication.
- Implemented secure authentication and authorization systems using **JWT** and **RBAC**.
- Containerized backend applications using **Docker** and configured reverse proxy services using **Nginx**.
- Integrated **MySQL** and **MS SQL** databases for production data pipelines and enterprise workflows.

Technical Skills

Architecture: Microservices, Monolithic, MVC, MVVM, Event-Driven Systems
Backend & Python: Python (FastAPI, Django), Dart, OOP, Async/Await, Pydantic, REST API Development, SQLAlchemy
Frontend: Flutter, Dart, HTML, CSS, Kivy, KivyMD
Databases: SQL (Queries, Joins, Indexing), MySQL, MS SQL, PostgreSQL, MongoDB
DevOps & Security: Docker, Nginx, CI/CD (GitHub Actions), OAuth 2.0, JWT, RBAC, RSA Encryption
Connectivity: WebSockets, MQTT, Redis, TCP/IP, Serial Communication (RS232/RS485)
Tools: Git, GitHub, PyCharm, Android Studio, VS Code

Projects

Industrial IoT & Automation (ChemChef) | *Flutter, FastAPI, Dart, TCP/IP, RS-232/485, MS SQL, MySQL, RSA, WebSockets*

- Developed a desktop SCADA automation system for automated chemical dosing, interfacing directly with plant scales and PLCs via **TCP/IP** and **RS-232/RS-485 serial** with CRC verification.
- Implemented an offline-resilient **MySQL Proxy** / **Failover Router** synchronizing batch recipes with enterprise **MS SQL ERP**, streaming real-time telemetry via **WebSockets** and securing payloads with **RSA encryption**.

Enterprise Lifecycle Manager (ERP/MES) | *Flutter, FastAPI, Microservices, MySQL, Redis, Docker, Nginx, GitHub Actions, RBAC, JWT*

- Developed modular **FastAPI microservices** behind an **Nginx** reverse proxy to manage inventory levels, production work orders, machine licensing, and employee task assignments.
- Implemented **Redis** for session caching, **Alembic** for database migrations, and granular **RBAC/JWT** authorization, automating build and test workflows via **GitHub Actions**.

SolarOps – Industrial IoT Monitoring | *Flutter, FastAPI, MQTT, Monolithic, MySQL, Redis, Docker, Nginx, GitHub Actions, WebSockets*

- Engineered an IoT monitoring dashboard combining a centralized backend for real-time machine command execution with dedicated **microservices** for customer account and fleet management.
- Integrated **MQTT** brokers and **WebSockets** to process live sensor telemetry and dispatch control commands to Raspberry Pi units with sub-second latency, deploying via **Docker** and **Nginx**.

ServiceLink (Enterprise SaaS ERP) | *Flutter, FastAPI, Microservices, MySQL, Redis, WebSockets, GitHub Actions, macOS Deployment*

- Built a cross-platform service management application for **macOS** and Web, orchestrating **FastAPI microservices** for customer product registries, complaint logging, and invoice generation.
- Implemented real-time ticket dispatching and status updates using **WebSockets** and **Redis pub/sub**, enabling instant coordination across field engineers, managers, and billing teams.